

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
ACADEMIC YEAR 2025-2026 EVEN SEMSTER

SUBJECT CODE: AL3451

SUBJECT NAME: MACHINE LEARNING

YEAR/SEM: II/IV

UNIT I - INTRODUCTION TO MACHINE LEARNING			CO
Review of Linear Algebra for machine learning; Introduction and motivation for machine learning; Examples of machine learning applications, Vapnik-Chervonenkis (VC) dimension, Probably Approximately Correct (PAC) learning, Hypothesis spaces, Inductive bias, Generalization, Bias variance trade-off.			CO 1
Q. No.	Questions PART A	Marks	Knowledge Level
1.	What is a vector, matrix and Eigen vector? How is it applied in machine learning?	2	K2
2.	What does it mean for a matrix to be singular?	2	K2
3.	Define the concept of orthogonal vectors and their relevance in ML.	2	K1
4.	Define machine learning? List the classification of machine learning algorithms.	2	K1
5.	Differentiate between supervised and unsupervised learning.	2	K2
6.	What is reinforcement learning, and how does it differ from supervised learning?	2	K2
7.	What is overfitting and under fitting in machine learning?	2	K2
8.	What is candidate elimination problem?	2	K2
9.	What is PAC learning? What are the parameters of PAC learning?	2	K2
10.	What do you mean by hypothesis space?	2	K2

11.	What is inductive bias in ML?	2	K2						
12.	Define concept learning.	2	K1						
13.	For a given sample dataset<table><tr><th>Name</th><th>Age</th></tr><tr><td>Anish</td><td>38</td></tr><tr><td>Vino</td><td>25</td></tr></table> Find the number of instance space, hypothesis space and Semantically distinct hypothesis.	Name	Age	Anish	38	Vino	25	2	K2
Name	Age								
Anish	38								
Vino	25								
14.	When a hypothesis is said to be consistent? Give an example	2	K2						
15.	Differentiate maximally Specific hypothesis and minimally specific hypothesis.	2	K2						

Q. No.	Questions PART B	Marks	Knowl edge Level
1.	Illustrate the different ways to represent the data in linear algebra and its operations.	13	K2
2.	Describe the applications of Machine Learning in various domain	13	K2
3.	Elucidate the Vapnik-Chervonenkis (VC) dimension with suitable examples	13	K2
4.	Elaborate on PAC learning Give an example.	13	K2
5.	Discuss the importance of hypothesis spaces in machine learning and how they impact learning performance	13	K2
6.	Explain in detail about the inductive bias and Bias- Variance trade-off with suitable sketch	13	K2

UNIT II SUPERVISED LEARNING			CO
Linear Regression Models: Least squares, single & multiple variables, Bayesian linear regression, gradient descent, Linear Classification Models: Discriminant function – Perceptron algorithm, Probabilistic discriminative model - Logistic regression, Probabilistic generative model – Naïve Bayes, Maximum margin classifier – Support vector machine, Decision Tree, Random Forests			CO2
Q. No.	Questions PART A	Marks	Knowledge Level
1.	What is the primary goal of linear regression?	2	K2
2.	What distinguishes multiple linear regressions from simple linear regression?	2	K2
3.	Mention the merits of Bayesian linear regression	2	K2
4.	What is gradient? How gradient descent is useful in ML?	2	K2
5.	What is the discriminant function in the context of classification?	2	K2
6.	What is the key characteristic of logistic regression?	2	K2
7.	Compare the linear regression model and logistic regression model	2	K2
8.	How does a linear SVM differ from a non-linear SVM?	2	K2
9.	What is the significance of the regularization parameter (C) in SVM?	2	K2
10.	What is the role of the kernel function in SVM?	2	K2
11.	How does a decision tree classify data?	2	K2
12.	What is the Gini index, and how is it used in decision trees?	2	K2
13.	What is entropy and information gain in decision tree? Write the formula to find entropy and information gain.	2	K2

14.	What is the purpose of perceptron algorithm?					2	K2
Q. No.	Questions PART B					Marks	Knowledge Level
1.	By the method of least squares find the straight line to the data given below.					13	K3
	X	5	10	25	20	25	
	Y	16	19	23	26	30	
2.	Write a short notes on a) Regression and Correlation b) Limitations of regression models					6 7	K2
3.	Discuss Bayesian linear regression and its advantages over standard linear regression.					13	K2
4.	Explain the gradient descent algorithm and its application in optimizing linear regression models.					13	K2
5.	Consider a boy who has a volleyball tournament the next day but feels sick today. Unusually, there is only a 40% chance he would fall sick since he is a healthy boy. Now, Find the probability of the boy participating in the tournament. The boy is very interested in volleyball, so there is a 90% probability that he will participate in tournaments and 20% fall sick, given that he participates in the					15	K3
6.	Explain about how optimal hyperplane differ from other hyperplanes. Elaborate on how SVM is able to achieve this?					13 13	K2
7.	Explain the process of constructing CART (Classification and Regression Tree).					13	K3
8.	With an example explain the Decision Tree concepts in detail.					13	K3
9.	Consider the following training dataset used to decide whether					15	K3

to play tennis based on weather conditions.

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

1. Compute the entropy of the entire dataset.
2. Calculate the Information Gain for the attribute **Outlook**.
3. Determine the root node of the decision tree using the ID3 algorithm.
4. Construct the first level of the decision tree.
5. Continue the splitting process until the tree is fully constructed.

Use base-2 logarithm for entropy calculations.

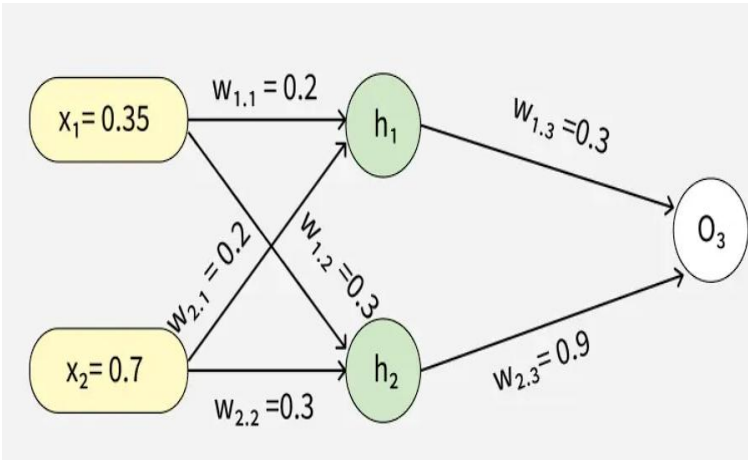
UNIT III ENSEMBLE TECHNIQUES AND UNSUPERVISED LEARNING			CO
Combining multiple learners: Model combination schemes, Voting, Ensemble Learning - bagging, boosting, stacking, Unsupervised learning: K-means, Instance Based Learning: KNN, Gaussian mixture models and Expectation maximization.			CO212.3
Q. No.	Questions PART A	Marks	Knowle dge Level
1.	What is meant by combining models? What are the schemes in model combination.	2	K2
2.	Define Voting.	2	K2
3.	Differentiate between Boosting, Bagging and stacking.	2	K2
4.	Compare Hard Voting vs. Soft Voting in ensemble learning.	2	K2
5.	Define bootstrapping and aggregation.	2	K2
6.	Define Expectation Maximization.		K2
7.	Define Centroid in K-Means clustering.	2	K2
8.	Identify the applications and challenges of clustering algorithm		K2
9.	Define Gaussian Mixture Model (GMM).	2	K2
10.	How does Expectation Maximization (EM) optimize GMM parameters?	2	K2

Q. No.	Questions PART B	Marks	Know ledge Level
1.	i) Explain Bagging with an example. ii) Compute the expected accuracy of an ensemble using Bagging, given three weak classifiers with accuracies of 60%, 65%, and 70%.	6 7	K3
2.	i) Describe the Boosting algorithm and its working. ii) Given a weak classifier with 55% accuracy, explain how Boosting might improve its performance over multiple iterations.	6 7	K3
3.	i) Explain Stacking ensemble learning with an example. ii) Given three base classifiers with accuracies of 75%, 80%, and 85%, discuss how Stacking can improve overall accuracy	13	K3
4.	Cluster the following eight points (with (x, y) representing locations) into three clusters using K-means clustering method. A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9).	13	K3
5.	Explain Expectation Maximization (EM) with suitable example	13	K2
6.	Consider a boy who has a volleyball tournament the next day but feels sick today. Unusually, there is only a 40% chance he would fall sick since he is a healthy boy. Now, Find the probability of the boy participating in the tournament. The boy is very interested in volleyball, so there is a 90% probability that he will participate in tournaments and 20% fall sick, given that he participates in the tournament.	15	K3

7.	Explain KNN algorithm and solve the given problem. Consider the training dataset given in the following table. Use Weighted k-NN and determine the class. Test instance (7.6, 60, 8) and K=3. <table><tr><th>S.NO</th><th>CGPA</th><th>ASSESS MENT</th><th>PROJECT SUBMITT ED</th><th>RESULT</th></tr><tr><td>1</td><td>9.2</td><td>85</td><td>8</td><td>PASS</td></tr><tr><td>2</td><td>8</td><td>80</td><td>7</td><td>PASS</td></tr><tr><td>3</td><td>8.5</td><td>81</td><td>8</td><td>PASS</td></tr><tr><td>4</td><td>6</td><td>45</td><td>5</td><td>FAIL</td></tr><tr><td>5</td><td>6.5</td><td>50</td><td>4</td><td>FAIL</td></tr><tr><td>6</td><td>8.2</td><td>72</td><td>7</td><td>PASS</td></tr><tr><td>7</td><td>5.8</td><td>38</td><td>5</td><td>FAIL</td></tr><tr><td>8</td><td>8.9</td><td>91</td><td>9</td><td>PASS</td></tr></table>	S.NO	CGPA	ASSESS MENT	PROJECT SUBMITT ED	RESULT	1	9.2	85	8	PASS	2	8	80	7	PASS	3	8.5	81	8	PASS	4	6	45	5	FAIL	5	6.5	50	4	FAIL	6	8.2	72	7	PASS	7	5.8	38	5	FAIL	8	8.9	91	9	PASS	15	K3
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UNIT IV - NEURAL NETWORKS			CO
Multilayer perceptron, activation functions, network training – gradient descent optimization – stochastic gradient descent, error backpropagation, from shallow networks to deep networks –Unit saturation (aka the vanishing gradient problem) – ReLU, hyperparameter tuning, batch normalization, regularization, dropout.			CO 4
Q. No.	Questions PART A	Marks	Knowledge Level
1.	List the types of activation functions used in neural networks.	2	K1
2.	Draw the structure of an artificial single neuron based on a biological neuron.	2	K1
3.	Give the types of ANN?	2	K1
4.	Distinguish between perceptron and Multi-layer perceptron.	2	K1
5.	Why could the XOR problem not be solved by a simple perceptron?	2	K2
6.	Define ReLU (Rectified Linear Unit) activation function.	2	K2
7.	Differentiate between the shallow and deep networks.	2	K2
8.	Compare gradient descent and stochastic gradient descent (SGD).	2	K2
9.	List the problems associated with back propagation neural network.	2	K1
10.	Differentiate between single layer and multi-layer perceptron.	2	K2
11.	Why is the sigmoid activation function not preferred in deep networks?	2	K2
12.	List the advantages of using ReLU over sigmoid and tanh.	2	K1
13.	How does L1 and L2 regularization work in neural networks?	2	K2
14.	Why is hyper parameter tuning important in deep learning?	2	K2

Q. No.	Questions PART B & C	Marks	Know ledge Level
1.	Explain the architecture of a multilayer perceptron (MLP) and its applications in machine learning.	13	K3
2.	Explain hyper parameter tuning with example.	13	K3
3.	Discuss in detail about back propagation concept in ANN.	13	K3
4.	Elaborate on Regularization. Drop-outs.	6 7	K2
5.	Compare shallow networks and deep networks in terms of representation power, training difficulty, and generalization.	8	K3
6.	Consider a dataset where neural networks overfit. Apply regularization techniques (L1, L2, dropout) to improve generalization.	13	K3
7.	What is batch normalization? Explain its role in stabilizing and accelerating neural network training.	13	K2
8.	Discuss the vanishing gradient problem in training deep networks and techniques to address it.	13	K2

Q. No.	Questions PART B & C	Marks	Knowledge Level
1.	A neural network consists of 2 input neurons, 2 hidden neurons, and 1 output neuron. Using the given weights, biases, and sigmoid activation, perform one complete backpropagation step. Clearly show the forward pass and backward pass computations. The target output is 0.5 and the learning rate is 1.  Perform one forward propagation and backward propagation.	15	K3
2.	Consider a neural network with one hidden layer using Sigmoid, Tanh, and ReLU activation functions respectively in three different experiments. - For a given input $x = -1, 0, 2$, compute the output of each activation function. - Plot or tabulate the outputs. - Analyze how the choice of activation function affects gradient propagation and learning behavior in the network.	15	

UNIT V DESIGN AND ANALYSIS OF MACHINE LEARNING EXPERIMENTS			CO
Guidelines for machine learning experiments, Cross Validation (CV) and resampling – K-fold CV, bootstrapping, measuring classifier performance, assessing a single classification algorithm and comparing two classification algorithms – t test, McNemar's test, K-fold CV paired t test			CO 5
Q. No.	Questions PART A	Marks	Knowledge Level
1.	Define cross-validation. What are the methods of cross validation?	2	K2
2.	List the various phases of machine learning life cycle.	2	K1
3.	List different benefits of cross-validation techniques .	2	K1
4.	Write the steps involved in conducting a K-fold CV experiment.	2	K2
5.	State the Confusion Matrix.	2	K2
6.	What is the purpose of the t-test in machine learning experiments?	2	K2
7.	What is bootstrapping sampling?	2	K2
8.	What are Type I and Type II errors in hypothesis testing?	2	K2
9.	What is the role of statistical significance in comparing classifiers?	2	K2
10.	What is log loss?	2	K2
11.	List the performance evaluation measures.	2	K1
12.	Differentiate between K-fold CV and Leave-One-Out CV.	2	K2
13.	Why is bootstrapping useful in small datasets?	2	K2
14.	Differentiate between parametric and non-parametric tests.	2	K2

Q. No.	Questions PART B & C	Marks	Knowledge Level
1	With neat diagram explain about K-fold Cross Validation technique.	13	K2
2.	Explain the various methods of measuring Classifier Performance with suitable examples.	13	K2
3	Elaborate the t test, McNemar's test and K-fold CV paired t test by giving your own example.	13	K3
4	List and illuminate the guidelines for Machine Learning Experiments.	13	K3
5	A bank is developing a fraud detection model using a dataset where fraud cases constitute only 0.17% of total transactions. Design a complete evaluation framework to ensure fair comparison between two classification models.	15	K3